DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

Eigen Values and CHT

Lecture No. 03



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Recap of previous lecture



Topic Eigen Values

Topics to be Covered



Topic

1. Cayley Hamilton Theorem
2. L-U Decomposition.

Cayley-Hamilton Theorem → "Every Square Mat satisfies it's own C.Eq"

ie we can Replace $(\lambda \longrightarrow A)$ in CEq.

for eg if $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ then it's CEqu^n is $|A - \lambda I| = 0$

$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0 \longrightarrow$ CEq.

By C-H-Th, $(\lambda \to A)$

$\hookrightarrow A^3 - 5A^2 + 7A - 3I = O$

In general if $A_{n\times n}$ then it's CEq is $|A-\lambda I|=0$

or $\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \dots + a_{n-1}\lambda + a_n = 0$

By C.H.T, $\lambda \to A$

$$1.A^n + a_1A^{n-1} + a_2A^{n-2} + \dots + a_{n-1}A + a_nI = O$$

Application: with the help of C.H.Th we can find $Tr(A)$, $|A|$ & $\bar{A}^{-1}$ as follows;

① $Tr(A) = -(a_1)$

② $|A| = (-1)^n a_n$

③ $\bar{A}^{-1}$ = will be discussed in Q.

Take Care: → these Results are applicable when Coeff of highest Power of A should be unity.

(4) Shortcut Method of writing C.Equⁿ of $A_{2\times 2}$

C.Equⁿ is $|A-\lambda I|=0$

or $\lambda^2+a_1\lambda+a_2=0$

$\lambda^2-(-a_1)\lambda+((-1)^2 a_2)=0$

$$\boxed{\lambda^2-(\text{Tr}\,A)\lambda+(|A|)=0}$$

eg For $A=\begin{bmatrix}4 & 2\\ 2 & 4\end{bmatrix}$, C.Equⁿ:

$\lambda^2-(8)\lambda+(12)=0$

(5) Constant term in the C.Equⁿ of $A_{n\times n}=\boxed{a_n=(-1)^n|A|}=\begin{cases}|A|, & n=\text{even}\\ -|A|, & n=\text{odd}\end{cases}$

Proof: we know that $|A|=(-1)^n a_n$

or $(-1)^n a_n=|A|$

$(-1)^n(-1)^n a_n=(-1)^n|A|$

$(-1)^{2n} a_n=(-1)^n|A|$

$a_n=(-1)^n|A|$ (Hence Proved)

Verification: if $A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}_{3\times3}$ then C-Eqn $\lambda^3-5\lambda^2+7\lambda-3=0$

& By C-H Th, $\lambda \to A$ i.e. $\boxed{A^3-5A^2+7A-3I=O}$ —— (1)

on Comparision with, $A^3+a_1A^2+a_2A+a_3I=O$

$\Rightarrow a_1=-5,\ a_2=7,\ a_3=-3$

Now $Tr(A)=-(a_1)=-(-5)=+5$

& $|A|=(-1)^3 a_3=-(-3)=+3$

Now $\because |A|\neq 0 \Rightarrow \bar{A}^1$ will exist.

Pre Multiply by $\bar{A}^1$ in (1)

$\bar{A}^1(A^3-5A^2+7A-3I)=\bar{A}^1\cdot O$

$A^2-5A+7I-3\bar{A}^1=O$

$-3\bar{A}^1=-(A^2-5A+7I)$

$\bar{A}^1=\frac{1}{3}(A^2-5A+7I)$ Ans

(iv) $\text{adj}\,A=A^2-5A+7I$

(v) Also Evaluate $B = A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 7A^2 - 2A + 4I$ & $|B| = ?$

we know that By (1)

$$A^3 - 5A^2 + 7A - 3I = O \quad \text{---(1)}$$

$$B = A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + A + 4I$$

$$= A^5(O) + A(O) + A + 4I$$

$$B = A + 4I \quad \text{Ans.}$$

$$= \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} + 4\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 1 & 1 \\ 1 & 6 & 1 \\ 0 & 0 & 5 \end{bmatrix}$$

(vi) $|B| = 0 - 0 + 5[36 - 1] = 175$ Ans

P8 If $P_{3\times3}$ s.t its C.Eq is $a(\lambda)=|\lambda I-P|=\lambda^3+\lambda^2+\lambda+1=0$ $=f(\lambda)$

Gate then find $Tr(P)$, $|P|$, P^{-1}

Sol: C.Equ of P is $|P-\lambda I|=0$ or $|\lambda I-P|=0$ or $\lambda^3+\lambda^2+\lambda+1=0$

By C.H.Th, $\lambda \to P$ in C.Eq. ie we will get $P^3+P^2+P+1.I=0$ ——— (1)

on Comparision with $(P^3+a_1P^2+a_2P+a_3I=0)$

(i) $Tr(P)=-(a_1)=-(1)=-1$

(ii) $|P|=(-1)^3 a_3=-(1)=-1$

Note- $|P-\lambda I|=0$

$(-1)^3|\lambda I-P|=0$

$|\lambda I-P|=\frac{0}{-1}=0$

(iii) By (1), $P^{-1}(P^3+P^2+P+I)=P^{-1}.0$

$P^2+P+I+P^{-1}=0$

$P^{-1}=-(P^2+P+I)$ Ans

(iv) Also Evaluate $|B|$ if $B = P^8 + P^7 + P^6 + P^5 + P^4 + P^3 + P^2 + P + 4I$

we have

$P^3 + P^2 + P + I = O$ ——— (1)

$B = P^5(P^3 + P^2 + P + I) + P(P^3 + P^2 + P + I) + 4I$

$= P^5(O) + P(O) + 4I$

$B = 4I$

$\therefore |B| = |4I|_{3\times 3} = 4^3 |I| = 64 \times 1 = 64$

(a) 4

(b) 1

(c) 0

(d) ✓ 64

Q: if $A = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{4\times4}$ s.t it's E.Values are $1, -1, i, -i$ then $A^4 = ?$

(a) 1

(b) i

(c) I_4

(d) 0

given that $\lambda = 1, -1, i, -i$

so C.Equ^n of A is

$$(\lambda-1)(\lambda+1)(\lambda-i)(\lambda+i)=0$$

$$(\lambda^2-1)(\lambda^2+1)=0$$

$$\boxed{\lambda^4-1=0}$$

By C-H-T $(\lambda \to A)$, $A^4 - I = 0 \Rightarrow A^4 = I = I_{4\times4}$

$|A| = (1)(-1)(i)(-i)$

$= (-1)(-i^2)$

$= -1$

= constant term.

Q: If $A=\begin{bmatrix}-1 & -2\\ 0 & 3\end{bmatrix}$ then A will satisfy

MSQ

(a) $A^2-2A-3I$

(b) $A-2I-3A^{-1}$

(c) $(A+I)(A-3I)=0$

(d) None

Sol: C.Eq^n of A is $\lambda^2-(\text{tr}(A))\lambda+(|A|)=0$

$\lambda^2-(2)\lambda+(-3)=0$

By C.H.Th, $\boxed{A^2-2A-3I=0}$ Ans.

Pre Multiply by A^{-1}

$\boxed{A-2I-3A^{-1}=0}$ Ans

Factor

$\boxed{(A+I)(A-3I)=0}$ Ans

Q. If $A_{4\times4}$ s.t A satisfies $3A^4-6A^3+2A^2+4A+9I=O$ then find $|A|=?$

(M-I) $|A|=(-1)^4 a_4=(+1)(9)=9$ WRONG.

(M-II) we have $1.A^4-\frac{6}{3}A^3+\frac{2}{3}A^2+\frac{4}{3}A+\frac{9}{3}I=\frac{1}{3}(O)$

$1.A^4-2A^3+\frac{2}{3}A^2+\frac{4}{3}A+3I=O$

$a_1=-2,\ a_2=\frac{2}{3},\ a_3=\frac{4}{3},\ a_4=3.$

$\text{Tr}\,A=-(a_1)$
$=-(-2)$
$=+2$

$|A|=(-1)^4 a_4=(+1)(3)=3$ Ans

Q. the constant term in the C.Eq of $A=\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}_{4\times4}$ will be ?

Sol:

$$a_n = (-1)^4 |A| = |A| = \cdots = 5$$ Ans.

L-U Decomposition (L-U Factorisation)

We can decompose any square Mat into the product of L-T-M & U-T-M by using following 3 Methods,

① Dolittle Method

$$A = (\text{unit L-T-M})(\text{U-T-M})$$

$$A_{3\times3} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

② Crout Method

$$A = (\text{L-T-M})(\text{unit U-T-M})$$

$$= \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

③ CHOLESKY Method

$$A = LL^{\theta}$$

$$= (LTM)\left(\overline{LTM}\right)^{T}$$

Q. Convert the following Mat $A=\begin{bmatrix}1 & 2\\ 2 & 13\end{bmatrix}$ into the product of LTM & U.T.M by all 3 Methods.

Sol: Dolittle Method →

$$A = (\text{unit LTM})(\text{UTM})$$

$$\begin{bmatrix}1 & 2\\ 2 & 13\end{bmatrix} = \begin{bmatrix}1 & 0\\ l_{21} & 1\end{bmatrix}\begin{bmatrix}u_{11} & u_{12}\\ 0 & u_{22}\end{bmatrix} \quad \text{—(1)}$$

$$\begin{bmatrix}1 & 2\\ 2 & 13\end{bmatrix} = \begin{bmatrix}u_{11} & u_{12}\\ l_{21}u_{11} & (l_{21}u_{12}+u_{22})\end{bmatrix}$$

$u_{11}=1$, $u_{12}=2$, $l_{21}u_{11}=2$, $l_{21}u_{12}+u_{22}=13$

$l_{21}=2$ $\quad (2)(2)+u_{22}=13$

$u_{22}=9$.

By (1) $A=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & 9\end{bmatrix}$

CROUT Method:

$A = (LTM)(Unit\ U.T.M)$

$$\begin{bmatrix} 1 & 2 \\ 2 & 13 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix} \begin{bmatrix} 1 & u_{12} \\ 0 & 1 \end{bmatrix} \quad \text{---} (1)$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 13 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{11}u_{12} \\ l_{21} & l_{21}u_{12} + l_{22} \end{bmatrix}$$

$l_{11} = 1$, $l_{21} = 2$, $l_{11}u_{12} = 2 \Rightarrow u_{12} = 2$

$l_{21}u_{12} + l_{22} = 13$

$(2)(2) + l_{22} = 13 \Rightarrow l_{22} = 9$

By (1),

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

CHOLESKY Method → $A=\begin{bmatrix}1 & 2\\ 2 & 13\end{bmatrix}$

Now $A=LL^{\theta}$

$\because$ A is Real so will L also.

$\hookrightarrow A=LL^{T}$

$$\begin{bmatrix}1 & 2\\ 2 & 13\end{bmatrix}=\begin{bmatrix}a & 0\\ b & c\end{bmatrix}\begin{bmatrix}a & b\\ 0 & c\end{bmatrix} \rightarrow (1)$$

$$\begin{bmatrix}1 & 2\\ 2 & 13\end{bmatrix}=\begin{bmatrix}a^2 & ab\\ ba & b^2+c^2\end{bmatrix}$$

$a^2=1,\ ab=2,\ b^2+c^2=13$

$\Rightarrow a=+1 \quad b=2,\ 4+c^2=13 \Rightarrow c=+3$

By (1), $A=\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & 3\end{bmatrix}$

Doolittle: $A=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & 9\end{bmatrix}$

Crout: $A=\begin{bmatrix}1 & 0\\ 2 & 9\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$

Application :- It is useful to solve systems of Equations

Consider $AX = B \Rightarrow X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = ?$ ——— (1)

we can write $(LU)X = B$

$L(UX) = B$

Let $UX = Y$ ——— (2) then $LY = B$ ——— (3)

By (3)

$$\begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$\Rightarrow y_1 = ?,\ y_2 = ?,\ y_3 = ?$

Hence $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ = Calculated

By (2), $UX = Y$

$$\begin{bmatrix} d & e & f \\ 0 & g & h \\ 0 & 0 & i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$x_3 = ?,\ x_2 = ?,\ x_1 = ?$

So $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ = Ans

THANK - YOU